

Thesis Defense Questions and Answers

How to Use This File

This file is designed for viva/thesis defense preparation based on:

- `Research_Proposal_Revised_Compressed.docx`
- `thesis_draft.md`
- supporting repository materials such as `final_thesis_summary.md` and visible code files

Each question has two answer versions:

- **Short answer:** at most 2 sentences, suitable for quick viva responses.
- **Long answer:** a more complete version for follow-up discussion.

Important note: where the proposal, summary, and code disagree, these answers are written to stay **defensible and evidence-based**. In a defense, it is safer to answer honestly from the implemented work than to overclaim.

1. General Introduction and Motivation

Q1. What is the title of your thesis?

Short answer:

My thesis is titled **Correlation-Adaptive Weather Graph Modeling for Complex Terrain**. Its broader motivation is improving weather-informed agricultural decision support in mountainous regions.

Long answer:

The thesis focuses on correlation-adaptive weather graph modeling in complex terrain, especially mountain regions where nearby stations may experience very different weather. The larger motivation is agriculture, because crop recommendation systems depend heavily on reliable local weather information. My work concentrates on strengthening that forecasting backbone. So the title reflects both the graph-based forecasting method and its intended use in rugged topographies such as Nepal.

Q2. What is the main problem your thesis addresses?

Short answer:

The thesis addresses inaccurate local weather forecasting in complex terrain. In mountainous regions, geographical closeness does not always mean meteorological similarity.

Long answer:

The central problem is that standard spatial assumptions break down in mountain environments. Two stations can be geographically close but separated by ridges, valleys, altitude differences, and local circulation patterns, so their weather can differ substantially. If a forecasting model connects stations mainly by physical distance, it may learn misleading relationships. My thesis addresses this by redefining connectivity using historical weather similarity instead of map distance alone.

Q3. Why is this problem important?

Short answer:

It is important because local weather strongly affects agriculture, planning, and hazard awareness. Errors in local forecasting can lead to poor decisions in complex terrain.

Long answer:

Accurate local weather forecasting matters for crop planning, irrigation timing, harvesting, and broader risk management. In mountainous regions, microclimates are common, so coarse or distance-based approaches can produce misleading local estimates. That becomes especially problematic when decisions are sensitive to rainfall, humidity, or temperature extremes. Improving local forecasting therefore has both scientific and practical value.

Q4. Why did you choose mountainous or complex terrain as your study context?

Short answer:

I chose complex terrain because it is where traditional distance-based assumptions fail most clearly. It provides a strong test case for whether functional connectivity is better than physical proximity.

Long answer:

Complex terrain creates strong local variation due to altitude, slope orientation, valleys, ridges, and rain-shadow effects. These conditions make it difficult for simple nearest-station or distance-based methods to represent actual weather behavior. That makes mountainous regions scientifically interesting and practically important. If a correlation-based graph is useful anywhere, it should be useful in such terrain.

Q5. What is your core research idea in one sentence?

Short answer:

My core idea is that weather stations should be connected by **historical signal similarity** rather than only by geographical distance.

Long answer:

The central idea is to replace a purely distance-based graph with a correlation-based graph built from historical weather behavior. Instead of asking which station is nearest, I ask which station behaves most similarly over time. This allows the graph to capture functional connections that geography alone may miss. That is the methodological shift at the heart of the thesis.

Q6. What is the novelty of your work?

Short answer:

The novelty lies in using a correlation-adaptive graph for station connectivity in complex terrain. This shifts the graph definition from physical proximity to functional weather similarity.

Long answer:

The most defensible novelty is the graph-construction strategy. In the implemented pipeline, station edges are weighted using the absolute Pearson correlation of historical **T2M** time series, then used within an STGCN for forecasting. This is meaningful because it directly targets the weakness of distance-based assumptions in mountainous environments. The contribution is not simply using a GNN, but redefining the graph itself around observed atmospheric similarity.

Q7. What is the practical motivation behind this work?

Short answer:

The practical motivation is to improve weather inputs for agricultural and local decision-support systems. Better weather forecasts can support more reliable crop-related recommendations.

Long answer:

The project is motivated by the fact that weather is a critical input to decision-support systems, especially in agriculture. If the weather estimate is wrong at the local level, downstream recommendations can also be wrong. By improving how weather stations are related inside the model, the forecasting backbone can become more realistic for complex terrain. That creates a stronger foundation for future crop recommendation tools.

2. Research Gap, Objectives, and Hypothesis

Q8. What research gap did you identify?

Short answer:

The gap is that many spatial forecasting approaches assume near locations are more similar than distant ones. In complex terrain, that assumption is often weak or false.

Long answer:

Many spatio-temporal forecasting models use graphs based on geodesic distance, inverse distance weighting, or related proximity rules. Those approaches work better when space is relatively smooth, but mountainous regions violate that assumption. The gap is the lack of a connectivity definition that reflects real meteorological similarity rather than only geographic closeness. My thesis addresses that gap through correlation-based graph construction.

Q9. What are the main objectives of your thesis?

Short answer:

My main objectives are to construct a correlation-based weather graph and use it in an STGCN for multi-variable forecasting. I also aim to show why this idea is especially relevant in complex terrain.

Long answer:

The first objective is to define station connectivity using historical weather similarity, specifically absolute Pearson correlation of station **T2M** series. The second objective is to integrate that graph into a spatio-temporal graph convolutional framework. The third objective is to forecast agriculturally relevant variables such as minimum temperature, relative humidity, and precipitation. A broader objective is to position the method as enabling infrastructure for future agricultural advisory systems.

Q10. What is your main hypothesis?

Short answer:

My main hypothesis is that a functionally connected weather graph will better represent complex terrain than a purely distance-based graph. Therefore, it should provide a better forecasting structure for local weather modeling.

Long answer:

The hypothesis is that historical co-behavior between stations is a more informative basis for graph connectivity than physical distance alone in complex terrain. In mountainous regions, topographic barriers can separate nearby locations while distant valleys may share similar weather regimes. A correlation-based graph is intended to discover those hidden relationships. If that graph better reflects reality, the downstream forecasting model should also become more meaningful.

Q11. Did you fully prove superiority over a baseline?

Short answer:

Not fully within the repository evidence. The strongest supported claim is that the correlation-based forecasting pipeline is implemented, trainable, and scientifically justified.

Long answer:

This is an important point to answer honestly. The repository clearly supports the implemented correlation-based pipeline, including graph construction, model training, and sample inference outputs. However, it does not contain a fully documented, directly comparable 320-station baseline experiment with matching evaluation tables. So I present the work as a strong implemented method with a clear scientific rationale, while treating full comparative proof as incomplete or future work unless additional validated results are available elsewhere.

Q12. How would you summarize your thesis contribution in one paragraph?

Short answer:

My thesis contributes an evidence-based weather forecasting pipeline for complex terrain that builds station graphs from historical correlation instead of physical distance. It shows how this graph can be used within an STGCN to forecast key weather variables at station level.

Long answer:

The thesis contributes a correlation-adaptive graph modeling approach for station-level weather forecasting in complex terrain. The implemented system constructs graph edges from absolute Pearson correlation over historical **T2M** time series, sparsifies the graph, and then uses it inside a spatio-temporal graph convolutional network. The model is configured to forecast **T2M_MIN**, **RH2M**, and **PRECTOTCORR** from 43 input lags over a seven-step horizon. The broader significance is that it reframes station connectivity as functional rather than merely geographic.

3. Dataset and Study Inputs

Q13. What data did you use in this thesis?

Short answer:

I used weather-station data organized around 320 stations, along with processed graph and test artifacts in the repository. The main visible historical archive is **Weather_320_Stations_DB.csv**.

Long answer:

The repository shows a 320-station workflow supported by multiple files. The most important visible archive is **Weather_320_Stations_DB.csv**, which contains multi-date weather records and is consistent

with the temporal correlation logic. There is also `Stations_320_Final.csv`, but in the current repository snapshot it appears to be a single-date station snapshot rather than a full history. In addition, the test bundle includes graph tensors, station metadata, normalization statistics, and example test inputs.

Q14. Why did you choose 320 stations?

Short answer:

The 320-station dataset is the active implemented setting in the repository. It provides broad spatial coverage while remaining computationally manageable for graph-based modeling.

Long answer:

The 320-station setup is the one most clearly implemented and evidenced in the repository. It offers a substantial node set for graph learning while keeping the processing and model size tractable. It also appears to match the station metadata, processed graph files, and inference scripts that are available. So for a defensible thesis narrative, 320 stations is the most appropriate focus.

Q15. What variables are included in your model inputs?

Short answer:

The active 320-station pipeline uses 14 weather features, including temperature, humidity, precipitation, pressure, and wind variables. These include `T2M`, `T2MWET`, `TS`, `T2M_MAX`, `T2M_MIN`, `RH2M`, `PRECTOTCORR`, `PS`, and wind measurements.

Long answer:

The code shows 14 input features in the active pipeline: `T2M`, `T2MWET`, `TS`, `T2M_MAX`, `T2M_MIN`, `RH2M`, `PRECTOTCORR`, `PS`, `WS10M`, `WS10M_MAX`, `WS10M_MIN`, `WS50M`, `WS50M_MAX`, and `WS50M_MIN`. These features combine thermal, moisture, precipitation, pressure, and wind information. This gives the model a broader meteorological context than using one variable alone. However, the graph weights themselves are built specifically from `T2M` correlation.

Q16. What are the forecast target variables?

Short answer:

The implemented forecast targets are `T2M_MIN`, `RH2M`, and `PRECTOTCORR`. These correspond to minimum temperature, relative humidity, and corrected precipitation.

Long answer:

Although some narrative files mention slightly different targets, the inference script is the most reliable source for the active configuration. It explicitly maps the output labels to `T2M_MIN`, `RH2M`, and `PRECTOTCORR`. These variables are agriculturally meaningful because they relate to moisture availability, temperature stress, and rainfall conditions. Therefore, I treat those three as the operative targets in the thesis.

Q17. Why are these target variables important?

Short answer:

They are important because minimum temperature, humidity, and precipitation are key drivers of crop conditions and local weather risk. They are also directly forecast by the implemented model.

Long answer:

Minimum temperature matters for cold stress, humidity influences plant-water conditions and disease environments, and precipitation affects irrigation need and overall crop suitability. Together, these three variables provide a useful weather profile for downstream agricultural interpretation. They also represent distinct physical aspects of the atmosphere, making the forecasting task more meaningful than predicting only one variable. Most importantly, they are the outputs the code actually generates.

Q18. What is the difference between `Weather_320_Stations_DB.csv` and `Stations_320_Final.csv`?

Short answer:

`Weather_320_Stations_DB.csv` appears to be a multi-date historical archive, while `Stations_320_Final.csv` appears to be a one-row-per-station snapshot in the current repository. That difference matters because correlation over time requires historical series.

Long answer:

This is a key reproducibility issue. The graph-construction method needs time-series data so that each station has a historical signal that can be correlated with other stations. `Weather_320_Stations_DB.csv` is consistent with that need because it contains records across many dates. By contrast, the visible `Stations_320_Final.csv` looks like a station snapshot rather than a full temporal archive, so by itself it would not support the full correlation workflow.

Q19. What is the temporal setup of your model?

Short answer:

The model uses 43 input lags and predicts a 7-step forecast horizon. This is consistent across the active training and inference scripts.

Long answer:

The training script constructs the dataset with `lags=43` and `pred_seq=7`, and the inference script uses the same configuration. That means the model observes 43 past time steps and predicts the next 7 steps for the selected targets. This setup creates a short multi-step forecasting task rather than a single-step estimate. It also reflects the implemented configuration, so it is an evidence-based part of the thesis.

Q20. Did you do any normalization or preprocessing?

Short answer:

Yes, the visible pipeline computes global mean and standard deviation values for the selected features and uses them for normalization. The same statistics are reused during inference.

Long answer:

Normalization is important because the input variables are on very different scales, such as temperature, pressure, and precipitation. In the preprocessing path, mean and standard deviation values are computed for the selected features and saved to CSV files. During inference, the test data is normalized using those saved statistics, and predictions are later de-normalized for interpretability. This makes the training and test pipeline internally consistent.

4. Graph Construction and Mathematical Basis

Q21. What is a graph in the context of your thesis?

Short answer:

A graph is a network where weather stations are nodes and inter-station relationships are edges. The edge weights represent the strength of similarity between stations.

Long answer:

In my thesis, each weather station is represented as a node in a graph. The edges describe how strongly stations are connected, and those edge weights influence how information flows during graph convolution. Instead of defining those relationships only by geographic closeness, I use observed statistical similarity. This allows the graph to encode functional weather structure.

Q22. How did you construct the graph?

Short answer:

I constructed the graph by computing the absolute Pearson correlation between station **T2M** time series and using those values as edge weights. Then I retained only the strongest connections through sparsification.

Long answer:

The graph is built by first pivoting the weather table into a **Date x Station** matrix using **T2M** as the signal. Next, the Pearson correlation is computed between every pair of station columns, and the absolute value is taken so both strong positive and strong negative relationships are treated as strong connections in magnitude. These pairwise similarities become edge weights. Finally, the graph is sparsified by keeping only the top 33% of pairwise correlations and then adding symmetry and self-loops.

Q23. Why did you use Pearson correlation?

Short answer:

I used Pearson correlation because it measures how strongly two station time series co-vary over time. That makes it a reasonable first proxy for functional weather similarity.

Long answer:

Pearson correlation is simple, interpretable, and well suited for measuring linear co-variation between historical station signals. It provides a direct way to ask whether two stations tend to rise and fall together over time. In this work, it serves as a practical proxy for shared meteorological behavior. I chose it because it is mathematically transparent and easy to integrate into graph construction.

Q24. Why did you use the absolute value of Pearson correlation?

Short answer:

I used the absolute value so that strong anti-correlation is still treated as a meaningful relationship. This means the graph captures relationship strength, not just positive similarity.

Long answer:

If two stations are strongly anti-correlated, that still indicates a structured and informative relationship rather than no relationship. By taking the absolute value, the graph emphasizes magnitude of dependence

rather than sign. This is useful when the goal is to capture station coupling for message passing in the graph. However, one can also argue that sign information may contain physical meaning, so this is a design choice with trade-offs.

Q25. Can you write the edge-weight formula used in your graph?

Short answer:

Yes. The edge weight is defined as $(w_{ij} = |\rho(x_i, x_j)|)$, where (ρ) is the Pearson correlation between the historical T2M series of stations (i) and (j).

Long answer:

For stations (v_i) and (v_j) , I denote their historical T2M sequences as (x_i) and (x_j) . The Pearson correlation is computed as the covariance between the centered series divided by the product of their standard deviations. I then take the absolute value so the final graph weight is $(w_{ij} = |\rho(x_i, x_j)|)$. This gives a bounded similarity measure between 0 and 1 in magnitude after the absolute transformation.

Q26. Why did you use T2M to build the graph instead of all variables?

Short answer:

I used T2M as the primary signal because it is a strong general weather indicator and is explicitly used in the implemented code. It acts as a practical proxy for broader station similarity.

Long answer:

The implemented code constructs the graph from T2M, which is temperature at 2 meters. Temperature is often a stable and widely available signal that reflects broader atmospheric behavior and local terrain effects. Using one variable also keeps the graph definition simple and interpretable. A possible extension would be multi-signal or learned graph construction, but the current thesis uses T2M as the core similarity signal.

Q27. What does sparsification mean in your graph?

Short answer:

Sparsification means keeping only the strongest edges and removing weaker ones. This reduces noise and makes the graph more computationally efficient.

Long answer:

If every station is connected to every other station, the graph becomes dense and may include many weak or noisy relationships. Sparsification addresses this by retaining only stronger connections that are more likely to be meaningful. In my implementation, I keep the top 33% of pairwise correlation weights. This helps focus the model on the most relevant interactions while also lowering computational burden.

Q28. Why did you keep the top 33% of correlations?

Short answer:

The implementation uses a 33% retention rule to balance connectivity and noise reduction. It keeps the graph informative without making it fully dense.

Long answer:

The code explicitly sets `desired_ratio = 0.33`, meaning only the top third of pairwise correlations are

retained. This is a heuristic design choice that aims to preserve stronger relationships while discarding weaker ones that may mostly reflect noise. A denser graph could blur the most meaningful connections, while an overly sparse graph could lose useful structure. One limitation is that the repository does not show a formal ablation study justifying 33% as the optimum threshold.

Q29. Why are self-loops added to the graph?

Short answer:

Self-loops allow each node to retain its own information during graph convolution. They ensure a station's own history remains directly available to the model.

Long answer:

In graph neural networks, message passing combines information from neighbors. Without self-loops, a node's updated representation may rely too heavily on other nodes and lose direct access to its own features. By adding self-loops with weight 1, the model preserves self-information alongside neighbor information. This is a standard and useful graph-learning design choice.

Q30. How is your graph different from a geodesic-distance graph?

Short answer:

A geodesic-distance graph connects stations based on physical closeness, while my graph connects them based on historical weather similarity. So the difference is physical proximity versus functional connectivity.

Long answer:

In a distance-based graph, edge strength usually decreases as map distance increases, which assumes nearby stations are most informative. In my graph, edge strength is determined by observed co-behavior in historical data, regardless of whether stations are geographically near or relatively far apart. This allows the graph to capture valley-to-valley similarities or terrain-separated differences that geography alone may miss. That distinction is the central methodological contribution of the thesis.

Q31. What do you mean by "teleconnections" in your thesis?

Short answer:

I use "teleconnections" to describe meaningful weather relationships between stations that are not explained purely by geographic proximity. In practice, it refers to distant stations that behave similarly over time.

Long answer:

In the context of this thesis, teleconnections are functional links between stations whose time series show strong similarity even if they are not immediate spatial neighbors. The term emphasizes that atmospheric relationships can extend beyond simple local distance rules. I use it in a practical modeling sense rather than claiming a full climatological teleconnection analysis in the classical large-scale atmospheric literature. It helps explain why functional graphs may be more informative in complex terrain.

5. Model Architecture and Technical Design

Q32. What model did you use?

Short answer:

I used a Spatio-Temporal Graph Convolutional Network, or STGCN. It combines temporal sequence modeling with graph-based spatial learning.

Long answer:

The active implemented model is an STGCN, which is designed for graph-structured time series. It processes how conditions evolve over time while also learning how nodes influence one another through the graph structure. This makes it a natural choice for weather stations connected by learned inter-station relationships. In my thesis, the STGCN is the forecasting engine that operates on the correlation-derived graph.

Q33. Why is STGCN appropriate for your problem?

Short answer:

STGCN is appropriate because weather forecasting is both temporal and spatial. My problem requires learning from historical sequences across interacting stations, and STGCN is built for that.

Long answer:

Weather data has a strong time dimension because the past influences the future, and it also has a spatial or network dimension because nearby or behaviorally linked stations affect one another. STGCN explicitly combines temporal convolution with graph convolution, so it is suitable for this kind of structured forecasting problem. It is especially appropriate when the graph itself carries scientific meaning, as it does in my thesis. That is why it fits better than treating each station independently.

Q34. What is the active architecture in your implementation?

Short answer:

The active model is `STGCN_Best_BRC`, which uses three ST-Conv blocks followed by a linear layer. The block settings are `(14→128)`, `(128→64)`, `(64→16)` in the implemented architecture flow.

Long answer:

The training script instantiates `STGCN_Best_BRC`, so that is the architecture I treat as active. It uses three ST-Conv blocks with the following parameterization: `STConv(320, 14, 64, 128, 9, 4)`, `STConv(320, 128, 256, 64, 7, 4)`, and `STConv(320, 64, 32, 16, 5, 3)`, followed by `Linear(16, 3)`. This means the model gradually transforms the input through spatio-temporal graph operations before projecting to the three forecast targets. Since the code is the strongest evidence, I rely on that architecture rather than older summary prose.

Q35. What do the ST-Conv blocks do?

Short answer:

They jointly learn temporal patterns and graph-based spatial dependencies. In simple terms, they model how weather changes over time and how stations influence one another.

Long answer:

Each ST-Conv block applies temporal convolution to capture sequential patterns and graph convolution to propagate information across connected nodes. This means the model learns both the evolution of local weather and the influence of functionally linked stations. By stacking these blocks, the model builds

progressively richer spatio-temporal representations. That is what allows it to move beyond purely local or purely temporal forecasting.

Q36. Why did you not use a simple LSTM or CNN instead?

Short answer:

A simple LSTM or CNN can model temporal behavior, but it does not naturally encode inter-station graph structure. My problem specifically needs structured spatial relationships between stations.

Long answer:

An LSTM is strong for sequences and a CNN can capture local temporal patterns, but neither inherently models a learned station network in the same way a GNN does. My thesis is fundamentally about how station relationships should be represented in complex terrain. Because of that, the graph itself is not incidental but central to the research question. STGCN is therefore more aligned with the scientific objective than a purely sequential baseline.

Q37. Why did you predict three variables together instead of training three separate models?

Short answer:

Predicting them together allows the model to learn shared structure across related weather variables. It is also consistent with the implemented output layer of size 3.

Long answer:

Multi-output prediction can be beneficial when the targets are physically related, such as temperature, humidity, and precipitation. A shared model can learn common latent patterns that support all three outputs. It is also computationally more compact than training separate models for each variable. In my implementation, the final linear layer explicitly outputs three target channels, so the joint setup is part of the active design.

Q38. What libraries or frameworks did you use?

Short answer:

The implementation uses PyTorch and `torch_geometric_temporal`. Supporting processing also uses common Python tools such as pandas, NumPy, and NetworkX.

Long answer:

The core deep learning framework is PyTorch, and the spatio-temporal graph layers are implemented through `torch_geometric_temporal`, especially the `STConv` module. For data handling and preprocessing, the code uses pandas and NumPy. NetworkX is also used when constructing the station graph. Together, these tools provide the necessary support for graph creation, temporal data organization, model training, and inference.

6. Training, Validation, and Inference

Q39. How was the model trained?

Short answer:

The model was trained using temporal graph snapshots, Adam optimization, and early stopping. The active configuration uses 43 lags, 7 prediction steps, and a batch size of 32.

Long answer:

The training script loads precomputed snapshots and graph tensors, constructs a temporal graph dataset, and trains the STGCN using Adam. It uses a maximum of 300 epochs, an initial learning rate of 0.05, batch size 32, and early stopping with patience 15. The dataset is split into training and validation subsets. This provides a standard supervised forecasting setup over graph-structured time-series windows.

Q40. What optimizer and hyperparameters did you use?

Short answer:

I used the Adam optimizer with an initial learning rate of 0.05. The active setup also uses batch size 32, 43 lags, 7 forecast steps, and up to 300 epochs.

Long answer:

The visible training configuration uses Adam because it is robust and widely used for deep learning optimization. The initial learning rate is 0.05, with reduction when validation stops improving. The model is trained with a batch size of 32 for up to 300 epochs, though early stopping may end training sooner. These settings reflect the exact active script and are therefore the most reliable to report.

Q41. What kind of train-validation split did you use?

Short answer:

The training script uses an 85/15 split through `random_split`. So the validation set is created by random partitioning of the serialized temporal samples.

Long answer:

The active code uses `random_split` on the generated dataset rather than an explicitly chronological split. This means the data windows are randomly divided into training and validation subsets. It is straightforward to implement, but it has an important limitation for time-series forecasting because neighboring windows may share overlapping information. I would openly mention this as a validity consideration during the defense.

Q42. Why is `random_split` a limitation for time-series forecasting?

Short answer:

It can produce optimistic validation estimates because nearby temporal windows may overlap. A chronological split would better reflect real forecasting conditions.

Long answer:

In time-series forecasting, the ideal evaluation should preserve temporal order so the model is always tested on future data relative to training. With `random_split`, windows from similar periods may appear in both training and validation, especially when the windows overlap. That can reduce independence between the two sets and inflate performance estimates. So while the pipeline trains correctly, this validation strategy is weaker than a strict chronological holdout.

Q43. Did you use early stopping?

Short answer:

Yes, the training script uses early stopping with a patience of 15 epochs. This helps prevent unnecessary overtraining after validation improvement stalls.

Long answer:

Early stopping is implemented so training does not continue indefinitely after validation loss stops improving. In the script, patience is set to 15 epochs, and learning rate reduction is also applied before stopping. This is a practical regularization and efficiency measure. It is also supported by the training history logs in the repository.

Q44. What evidence do you have that the model trained successfully?

Short answer:

The repository contains training history logs showing substantial reduction in validation loss over time. It also includes saved checkpoints and example inference outputs.

Long answer:

The strongest direct evidence is the logged training history in `TEST/NW_WT/History.txt`. It shows both training and validation loss decreasing substantially from early epochs to the best saved checkpoint. The repository also preserves model checkpoints and example test predictions, which indicates that the training and inference pipeline was operational. While this is not the same as a complete benchmark study, it does support convergence and functionality.

Q45. What is your forecast horizon and what does it mean?

Short answer:

The forecast horizon is 7 steps. It means the model predicts the next 7 output time points after observing the 43-step input window.

Long answer:

Forecast horizon refers to how far ahead the model predicts into the future. In my implementation, after observing the previous 43 time steps, the model generates outputs for the next 7 steps for each target variable. This makes the task multi-step forecasting rather than one-step prediction. It is useful because many practical decisions benefit from short-range future outlooks rather than only the next immediate point.

Q46. How is inference performed for a new location?

Short answer:

The model forecasts at station nodes, and a user-supplied coordinate is mapped to the nearest available station. The final prediction is taken from that nearest station's forecast.

Long answer:

In the visible inference script, the model first produces forecasts for all stations in the graph. Then a new coordinate is provided, and geodesic distance is used to find the nearest station from the location metadata. The selected station's predicted values are returned as the practical forecast for that user location. So the application layer uses nearest-station retrieval after graph-based forecasting rather than a separate interpolation model.

Q47. Why does your final query stage still use nearest-station matching?

Short answer:

Because the model is trained at station nodes, the deployment step needs a practical way to map an external location to the existing graph. Nearest-station matching is a simple and usable solution for that purpose.

Long answer:

This is a useful distinction to clarify in the defense. The thesis improves the internal relational structure among stations using a correlation graph, but the final user query still has to connect a real coordinate to one of the modeled nodes. The current implementation does that by selecting the nearest station geographically. In future work, this could be improved with interpolation or learned spatial transfer, but for now it provides a practical node-selection mechanism.

7. Results, Interpretation, and Claims

Q48. What are the main results of your thesis?

Short answer:

The main supported results are that the correlation-based graph pipeline was implemented, trained, and used to produce multi-step forecasts. The repository also shows training convergence and sample inference outputs.

Long answer:

The strongest evidence-backed results are methodological and operational. The repository shows a complete correlation-based graph construction path, an STGCN training setup, training history with improving validation loss, and example multi-step forecasts. These demonstrate that the proposed forecasting framework is coherent and functional. What they do not yet fully provide is a complete comparative benchmark table strong enough to claim definitive superiority under a matched baseline protocol.

Q49. Did the validation loss improve during training?

Short answer:

Yes, the logs show a substantial reduction in validation loss from early training to the best checkpoint. This supports that the model learned meaningful patterns under the configured setup.

Long answer:

According to the visible history log, validation loss begins much higher in the initial epochs and declines significantly over training. The best saved checkpoint occurs much later in the run, indicating continued improvement before early stopping. This pattern suggests optimization convergence and nontrivial learning. It is one of the clearest empirical pieces of evidence in the repository.

Q50. Can you claim that your model is better than all other methods?

Short answer:

No, that would be too strong. The defensible claim is that my method is scientifically motivated,

implemented, and promising for complex terrain, but broader comparative validation remains limited in the visible repository.

Long answer:

In a defense, it is important not to overstate. I can confidently explain why correlation-based connectivity is a strong idea for mountainous weather modeling and show that the implemented system trains and produces forecasts. However, without a complete and reproducible baseline comparison across equivalent settings, I should not claim universal superiority. A more accurate claim is that the method is well justified, operational, and worthy of deeper comparative evaluation.

Q51. How do you interpret the sample forecasts?

Short answer:

I interpret them as evidence that the inference pipeline is operational and produces realistic-valued outputs. They demonstrate functionality, but not by themselves full predictive accuracy.

Long answer:

The sample forecasts show that the trained model can generate seven-step predictions for the target variables after normalization and de-normalization. The values appear physically plausible, which supports the technical validity of the inference pipeline. However, a sample prediction file is not enough to prove generalization performance unless accompanied by ground-truth comparison and systematic evaluation metrics. So I treat these outputs as operational evidence rather than final proof of superiority.

Q52. What is the scientific meaning of your result?

Short answer:

The scientific meaning is that weather-station connectivity in complex terrain may be represented more realistically through observed behavior than through distance alone. My thesis operationalizes that idea in a graph-learning framework.

Long answer:

The deeper contribution is conceptual as much as computational. The thesis argues that in rugged terrain, functional similarity can be a better organizing principle than geographic closeness for spatial learning. By turning that idea into a correlation graph and using it in an STGCN, the work provides a concrete way to test and apply that principle. Even where evaluation remains incomplete, the methodological argument is scientifically meaningful.

8. Limitations, Criticism, and Defense Questions

Q53. What are the main limitations of your thesis?

Short answer:

The main limitations are incomplete baseline comparison, reproducibility issues from hard-coded paths and missing intermediate artifacts, and a validation strategy that is not strictly chronological. These affect how strongly the results can be interpreted.

Long answer:

The first limitation is that the repository does not fully document a matched 320-station baseline

comparison with formal evaluation tables. The second is reproducibility, because some scripts depend on absolute Windows paths and some training artifacts are not included in portable form. A third limitation is the use of `random_split` for temporal data, which may overestimate validation performance. Finally, the crop recommendation stage is more of a downstream motivation than a fully implemented evaluated system in the current repository state.

Q54. Why is your thesis still valid despite these limitations?

Short answer:

It is still valid because the core methodological contribution is clearly implemented and scientifically coherent. The limitations mainly affect the strength of comparative and deployment claims, not the existence of the contribution itself.

Long answer:

Limitations do not erase the value of the work; they define its scope. My thesis still provides a clear graph-construction methodology, a mathematically grounded similarity measure, an implemented STGCN forecasting pipeline, and direct evidence of training and inference functionality. What the limitations do is narrow the claims I should make. So the thesis remains valid as a methodological and implementation contribution, while broader comparative and application claims should be framed more cautiously.

Q55. Why did you not include a stronger baseline comparison?

Short answer:

In the current repository evidence, a fully harmonized 320-station baseline is not available. I therefore chose to present the implemented contribution honestly rather than overstate unsupported comparisons.

Long answer:

The codebase includes traces of older location-driven preprocessing ideas, but not a fully aligned, logged, and directly comparable baseline experiment for the exact same 320-station setup. Since a thesis defense values credibility, I would rather acknowledge that gap than claim a comparison I cannot fully defend from the available evidence. This also highlights a strong direction for future work. In other words, the absence of a complete baseline is a limitation I explicitly recognize, not something I hide.

Q56. Why did you not use chronological validation?

Short answer:

The current script uses `random_split`, likely for implementation convenience and the existing serialized dataset structure. In hindsight, chronological validation would be more rigorous for forecasting.

Long answer:

The implemented workflow uses precomputed temporal samples and then applies `random_split`, which is straightforward in code. However, from a time-series evaluation perspective, a chronological split would better reflect real deployment conditions and reduce information leakage risk through overlapping windows. If challenged on this point, I would agree that chronological validation is the stronger protocol. I would position that as one of the first improvements for future experiments.

Q57. Could the correlation graph be sensitive to noisy data?

Short answer:

Yes, correlation estimates can be affected by noise, missingness, or nonstationarity in the historical series. That is one reason sparsification and careful preprocessing matter.

Long answer:

Correlation-based graphs depend directly on the quality and representativeness of the time-series data. If the underlying records are noisy, incomplete, or contain changing regimes, the estimated relationships may become less stable. Sparsifying the graph helps reduce weaker and potentially noisier connections, but it does not eliminate the issue entirely. A stronger future version could include robustness checks, rolling correlations, or dynamic graph learning.

Q58. Why did you not use a dynamic graph instead of a static one?

Short answer:

I used a static graph because it is simpler, interpretable, and supported by the current implementation. A dynamic graph is a strong future extension but was outside the present implemented scope.

Long answer:

The current thesis uses a static graph derived from historical relationships, which gives a stable and interpretable connectivity structure. This makes the methodology easier to explain and validate at the current stage. However, atmospheric relationships may vary by season or regime, so a dynamic graph could potentially capture changing dependencies more effectively. I would present dynamic graph learning as a natural next step rather than a weakness of the current scientific idea.

Q59. Does correlation imply causation in your graph?

Short answer:

No, correlation does not imply causation. In my thesis, correlation is used as a similarity measure for modeling, not as proof of physical causality.

Long answer:

This is an important conceptual distinction. The graph edges represent statistical association in historical time series, not a direct causal mechanism between stations. For forecasting, such associations can still be highly useful because they help identify informative relationships for message passing. But scientifically, I should be careful to say the graph captures functional similarity, not causal direction.

Q60. What if two stations are correlated only because of seasonality?

Short answer:

That is possible, and it is one reason the graph should be interpreted as a practical similarity structure rather than a perfect physical map. Seasonal effects can contribute to correlation strength.

Long answer:

Shared seasonal cycles can indeed inflate correlation between stations, even when their finer-scale local behaviors differ. That means the graph may mix broad climatological similarity with more localized coupling. In the current thesis, I treat the graph as a useful empirical representation rather than a perfect physical truth. A more advanced extension could separate seasonal components or build season-specific graphs.

Q61. Why did you not incorporate topography directly into the edge weights?

Short answer:

The current approach deliberately prioritizes observed weather behavior over predefined topographic heuristics. Topography is still relevant, but in this thesis it is treated more as context than as the main edge-weight driver.

Long answer:

Many prior methods already rely on geography, distance, or altitude rules, and my thesis intentionally shifts the focus toward data-driven functional similarity. The motivation is that topography matters precisely because it influences weather behavior, so capturing the behavior directly can be more informative than hard-coding the terrain relationship. That said, hybrid graphs combining correlation with elevation or terrain constraints could be a valuable future extension. I would frame that as an enhancement rather than a contradiction.

Q62. What is the biggest risk if someone overstates your thesis results?

Short answer:

The biggest risk is claiming stronger comparative or agricultural validation than the current evidence supports. That could weaken the credibility of the defense.

Long answer:

If I overclaim baseline superiority, complete crop recommendation success, or fully validated deployment readiness, an examiner can challenge those statements using the evidence gaps in the repository. It is much stronger to clearly separate what is implemented and supported from what is motivated or planned. My defense should therefore emphasize methodological contribution, operational forecasting evidence, and honest limitations. That makes the thesis more credible, not less.

9. Agricultural Relevance and Application

Q63. How does your work relate to crop recommendation?

Short answer:

My work provides the weather-forecasting backbone that a crop recommendation system can use. Better local weather inputs can lead to more reliable downstream agricultural advice.

Long answer:

Crop recommendation systems depend on environmental inputs such as temperature, humidity, and precipitation. My thesis focuses on improving those weather estimates in complex terrain, where poor local forecasting can degrade recommendation quality. So the relationship is upstream: I am improving the weather layer that would feed a crop advisory engine. In the current repository state, this agricultural role is best presented as enabling infrastructure rather than a fully implemented end-to-end recommender.

Q64. Did you build a full crop recommendation engine?

Short answer:

Not as a fully implemented and validated module in the visible repository. The thesis supports crop

recommendation mainly as motivation and future integration.

Long answer:

The current evidence strongly supports the weather-forecasting pipeline, but it does not show a complete crop-suitability engine with rules, thresholds, yield validation, and field evaluation. Therefore, I should not present crop recommendation as a finished evaluated subsystem unless I have separate validated materials to support that. Instead, I explain that the forecast variables are agriculturally relevant and can serve as inputs to a later advisory layer. This is a more accurate and defensible framing.

Q65. Why are minimum temperature, humidity, and precipitation useful for agriculture?

Short answer:

They influence crop stress, water availability, disease conditions, and planting suitability. These variables are therefore directly relevant to agricultural decisions.

Long answer:

Minimum temperature is important for frost or cold stress, especially in sensitive growth stages. Humidity influences evapotranspiration and disease-favorable environments, while precipitation affects soil moisture and irrigation demand. Together, these variables help characterize whether local conditions support or threaten crop performance. That is why they are meaningful targets for an agriculture-oriented forecasting system.

Q66. What would be needed to extend your thesis into a full crop advisory system?

Short answer:

It would need a crop-suitability model, agronomic rules or learned decision logic, and proper validation against crop outcomes. The weather forecasts alone are not enough.

Long answer:

A full crop advisory system would require at least three additional layers. First, it would need domain knowledge or a trained model linking weather and possibly soil data to crop suitability. Second, it would need a transparent decision framework that turns predicted weather into actionable recommendations. Third, it would need evaluation against agronomic references, expert judgments, or actual crop outcomes. My thesis creates a strong weather foundation for such a system, but does not yet complete that entire chain.

10. Methodology Justification and Examiner-Style Questions

Q67. Why is your approach better suited to Nepal-like terrain than standard interpolation?

Short answer:

Because Nepal-like terrain contains strong microclimates that are poorly captured by simple spatial smoothness assumptions. My approach lets the model learn from stations that behave similarly, not just from stations that are nearby.

Long answer:

Standard interpolation often assumes that weather changes smoothly with geographic distance. In rugged terrain, that assumption frequently breaks because valleys, ridges, and elevation differences create abrupt

local changes. My approach instead asks which stations show similar historical behavior, allowing the graph to connect functionally aligned places even when geography is misleading. This is why the method is particularly relevant for mountain regions.

Q68. Why did you choose a graph-based formulation instead of a grid-based one?

Short answer:

I chose a graph because weather stations are naturally irregularly distributed, not arranged on a regular grid. Graphs handle irregular node relationships much more naturally.

Long answer:

A grid-based model usually assumes data sampled on uniform spatial cells, but station networks are sparse and unevenly distributed. Graphs are better suited to such settings because they let us represent arbitrary node relationships and weighted connectivity. This flexibility is especially useful when the scientific question is itself about how connectivity should be defined. So the graph formulation is both practical and conceptually central.

Q69. Why not use geodesic distance and correlation together?

Short answer:

That is a valid extension and could create a hybrid graph. In this thesis, I used correlation alone to clearly test the idea of functional connectivity as the main organizing principle.

Long answer:

Combining distance and correlation could be a strong hybrid design because it would blend physical plausibility with data-driven similarity. However, for this thesis I wanted to isolate the effect of functional connectivity and show its conceptual importance in complex terrain. Using correlation alone also makes the core contribution easier to explain. I would describe a hybrid graph as one of the most promising directions for follow-up work.

Q70. Why did you use a static threshold instead of learning the graph end-to-end?

Short answer:

I used a fixed threshold because it is simple, interpretable, and directly supported by the current implementation. End-to-end graph learning is more flexible, but also more complex and less transparent.

Long answer:

For a thesis, interpretability matters. A fixed sparsification rule makes the graph construction transparent: the edges come directly from historical correlation and a known retention ratio. Learned graphs can be powerful, but they may also be harder to interpret and validate scientifically, especially in an early-stage system. I chose the simpler path to make the methodological contribution clearer and more defensible.

Q71. What assumptions does your method make?

Short answer:

It assumes historical **T2M** co-variation is a useful proxy for broader station similarity, and that a static correlation graph is informative for forecasting. It also assumes the available historical data is representative enough to build that graph.

Long answer:

The method assumes that stations with similar past temperature behavior are likely to share useful forecasting information. It also assumes that these relationships are stable enough over the modeling period to justify a static graph. Another assumption is that the data quality is adequate for estimating meaningful correlations. These assumptions are reasonable but should be stated openly, especially because they shape the interpretation of results.

Q72. If an examiner says your method is only an engineering trick, how will you respond?

Short answer:

I would say it is not just an engineering trick because it reflects a scientific hypothesis about how weather relationships should be modeled in complex terrain. The graph definition itself is the research contribution.

Long answer:

My response would be that the work is grounded in a clear scientific claim: physical proximity is not always the best proxy for meteorological similarity in mountainous environments. The choice to use correlation-based graph construction is therefore not arbitrary engineering, but an explicit model of functional connectivity. I then operationalize that hypothesis through a graph-learning framework. So the method is both computational and conceptually scientific.

Q73. If an examiner asks, "What is the single strongest part of your thesis?", what will you say?

Short answer:

The single strongest part is the clear and defensible shift from distance-based connectivity to correlation-based functional connectivity. It directly addresses the core weakness of weather modeling in complex terrain.

Long answer:

I would say the strongest part is the graph-construction idea itself. It is simple enough to explain mathematically, strong enough to be scientifically meaningful, and well aligned with the terrain-specific forecasting problem. It also clearly connects the motivation, methodology, and implementation. Even where evaluation can be strengthened, that conceptual core remains solid.

Q74. If an examiner asks, "What is the weakest part of your thesis?", what will you say?

Short answer:

The weakest part is the incomplete comparative evaluation and end-to-end application validation. The methodology is clearer and stronger than the current benchmark evidence.

Long answer:

I would answer honestly that the weakest part is the breadth of experimental validation. The repository does not fully support a rigorously matched 320-station baseline comparison, and the crop recommendation stage is not yet implemented as a complete validated subsystem. Those gaps limit how strong my final claims can be. However, I would immediately add that the methodological contribution itself remains clear and valuable.

11. Future Work and Conclusion Questions

Q75. What future work do you recommend?

Short answer:

I recommend chronological evaluation, stronger baseline benchmarking, dynamic or hybrid graph construction, and integration with a real crop recommendation module. These would strengthen both scientific rigor and practical impact.

Long answer:

The first priority is a more rigorous evaluation protocol, especially chronological train-validation-test splitting and matched baseline experiments. A second direction is graph refinement, such as dynamic graphs, hybrid correlation-topography graphs, or learned adjacency mechanisms. A third direction is practical extension into crop recommendation using agronomic rules or predictive suitability models. Together, these steps would turn the current forecasting framework into a more complete decision-support system.

Q76. How would you improve the thesis if you had more time?

Short answer:

I would add stronger ablation studies, a chronological evaluation setup, and a clearly documented baseline comparison. I would also make the full pipeline more reproducible and portable.

Long answer:

With more time, I would first improve reproducibility by removing hard-coded paths, packaging intermediate artifacts, and documenting the full workflow. Next, I would run controlled experiments comparing correlation-based, distance-based, and hybrid graphs under the same protocol. I would also test different sparsity thresholds and possibly seasonal or dynamic graph variants. Finally, I would connect the forecasts to a validated crop recommendation layer to strengthen the practical application story.

Q77. What is your final conclusion?

Short answer:

My final conclusion is that correlation-adaptive graph modeling is a promising and defensible approach for weather forecasting in complex terrain. It offers a more meaningful station-connectivity structure than relying on physical distance alone.

Long answer:

The thesis concludes that in complex terrain, representing station relationships through historical signal similarity is scientifically well motivated and practically useful for graph-based forecasting. The implemented pipeline demonstrates that this idea can be operationalized through a correlation-based graph and an STGCN forecasting model. The work is strongest as a methodological and forecasting contribution. Its broader comparative and agricultural claims should be advanced through future evaluation and integration work.

Q78. If you had to defend your thesis in one closing statement, what would you say?

Short answer:

I would say that my thesis shows why **functional connectivity** is a better modeling principle than **physical proximity alone** for weather forecasting in complex terrain. It turns that idea into an implemented, evidence-backed graph forecasting pipeline.

Long answer:

My closing statement would be that this thesis addresses a real weakness in local weather modeling for mountainous regions: the assumption that nearby stations are always the most informative. I replace that assumption with a correlation-adaptive graph that reflects how stations actually behave over time and integrate it into a spatio-temporal graph neural network. The result is a coherent, implemented forecasting framework with clear scientific motivation and practical relevance. Even where further validation is still needed, the core contribution is clear, defensible, and meaningful.

12. Quick Rapid-Fire Questions

Q79. What does STGCN stand for?

Short answer:

STGCN stands for **Spatio-Temporal Graph Convolutional Network**.

Long answer:

STGCN stands for Spatio-Temporal Graph Convolutional Network. It is a neural architecture designed for data that has both temporal sequence structure and graph-based spatial structure.

Q80. What does **T2M** mean?

Short answer:

T2M means air temperature measured at 2 meters above the surface.

Long answer:

T2M refers to temperature at 2 meters, which is a standard near-surface atmospheric measurement. In my thesis, it is the signal used to construct the inter-station correlation graph.

Q81. What does **RH2M** mean?

Short answer:

RH2M means relative humidity at 2 meters.

Long answer:

RH2M is relative humidity measured near the surface at 2 meters. It is one of the three forecast targets in the active implementation.

Q82. What does **PRECTOTCORR** mean?

Short answer:

PRECTOTCORR means corrected total precipitation.

Long answer:

PRECTOTCORR is a corrected precipitation variable and serves as one of the model outputs in the

implemented pipeline. It is especially important because precipitation is both difficult to predict and highly relevant to agriculture.

Q83. What is a node in your graph?

Short answer:

A node is a weather station.

Long answer:

In my graph, each node corresponds to one weather station. The node carries station-level feature values over time, and the graph edges describe how stations are related.

Q84. What is an edge in your graph?

Short answer:

An edge is the relationship between two stations. Its weight reflects the strength of their historical similarity.

Long answer:

An edge connects two station nodes in the graph. In my thesis, the edge weight is derived from the absolute Pearson correlation of their historical **T2M** signals.

Q85. What is the main takeaway in one line?

Short answer:

In complex terrain, **weather similarity matters more than map distance** when building a forecasting graph.

Long answer:

The one-line takeaway is that weather stations in complex terrain should be connected by how they behave, not only by where they are. That is the principle my thesis develops and implements.

Final Advice for Defense

- Answer from the **implemented evidence first**, especially when proposal claims are broader than the visible code.
- If asked about full crop recommendation or strong baseline superiority, respond honestly: **the forecasting backbone is implemented and supported; broader evaluation and downstream integration remain future work unless separately validated.**
- Keep repeating your central contrast clearly: **physical proximity vs functional connectivity.**
- If the examiner challenges limitations, acknowledge them directly and explain how they define scope rather than invalidate the contribution.